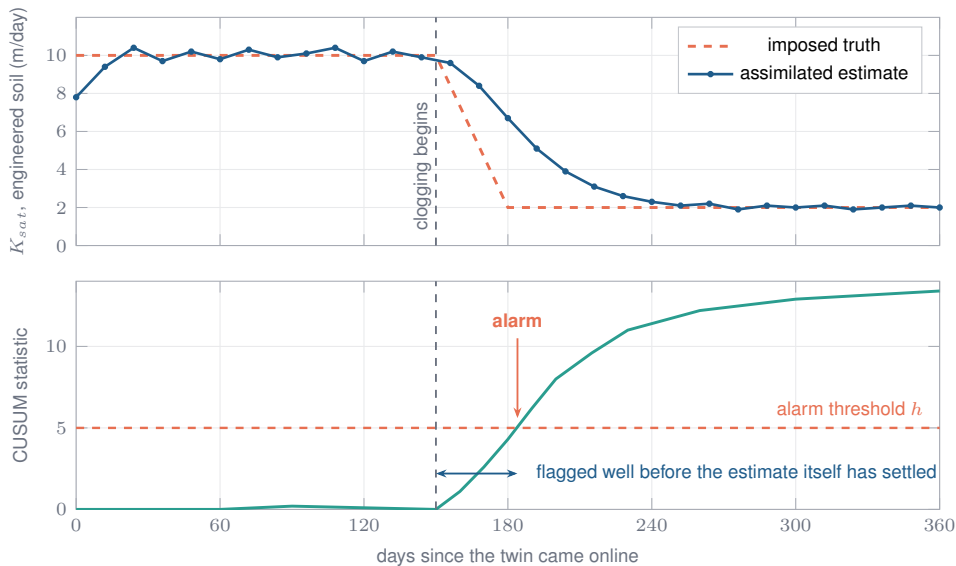




Because the twin re-estimates parameters continuously, the parameter trajectory is itself an **asset-condition signal**. A CUSUM statistic on the pooled cycle means, plus a Hotelling T^2 test over the whole parameter vector, turn that trajectory into an online alarm: the operator learns the medium is clogging while the estimate is still moving, not after it has converged on the new value.